BIBBOS

Business in a Box — Vessel Evolution

v5.1 → v6.3

Port of Spain, Trinidad and Tobago · February–March 2026

A record of one vessel's evolution — from a capable small business tool into a sovereign, self-replicating Business Operating System.

What This Document Is

This is a record of how BiBBOS grew — not just what it became, but how and why it grew the way it did. Written for colleagues, collaborators, and prospective stewards who want to understand the architecture, the decisions, and the distance travelled.

The vessel began as a capable single-file business tool. It became something that did not previously exist: a complete, self-teaching, self-replicating Business Operating System compressed into a single HTML file — sovereign, offline-capable, infinitely distributable, and free forever.

> *"Your business shouldn't pay monthly rent for its own brain."*

The Baseline: v5.1

v5.1 was already unusual. A single HTML file containing a functioning business tool — invoicing, purchase orders, basic inventory, customer and vendor management. Local storage only. No server. No account.

What it lacked was depth. The accounting was partial. HR did not exist. The intelligence layer was stub code. The vessel concept — the idea that each business is an isolated, portable, self-contained unit — was present but not fully realised. The file was a tool. It was not yet a system.

The Evolutionary Arc: v5.2 → v6.0

Phase 1 — Accounting Foundation (v5.2–v5.5)

The first major leap was moving from single-entry bookkeeping to a genuine double-entry accounting engine. Every transaction now posts balanced journal entries. Debits equal credits — always. The General Ledger tracks every account over time. The Balance Sheet equation (Assets = Liabilities + Equity) is computed live from the ledger, not from a simplified summary. The Income Statement derives from actual account activity.

Bank reconciliation followed: CSV import, journal entry matching, variance detection. The vessel became auditable.

Multi-currency support was added with secondary currency fields, live exchange rate wiring, and automatic Forex Gain/Loss account creation when a second currency is activated.

Phase 2 — HR Foundation (v5.6–v5.8)

The most complex structural addition. An HR module that handles the complete employee lifecycle inside the same file:

- Employee capsules with contract type, rate, NIS registration, banking details
- Timesheet entry and approval workflow
- Clock-in / clock-out punch tracking with QR badge scanning
- Payroll runs: gross calculation, statutory deductions, employer contributions, net pay
- Payslip generation with full deduction breakdown

- Payroll journal entries posted automatically to the General Ledger

The payroll accounting required careful attention to the Caribbean statutory landscape — PAYE annualisation logic, NIS employer vs employee splits, jurisdiction-aware rates throughout.

Phase 3 — Sales Depth (v5.9)

The sales layer expanded significantly:

- Quotes and Estimates with one-click conversion to invoices
- Recurring invoice engine (daily/weekly/monthly/bi-monthly/annual) with auto-generation on load
- Credit Notes with proper accounting reversal entries
- Encounter register for service businesses — client visits, sessions, billable events

Phase 4 — Intelligence and Delivery (v6.0 beta)

The intelligence layer moved from stub to functional:

- Revenue forecasting from historical invoice patterns
- Top customers and top products by revenue
- Cashflow insights — receivables vs payables position
- Inventory reorder alerts and margin tracking
- Delivery tracking with status progression and delivery note PDFs
- Marketing campaigns and customer touchpoints with sovereign broadcast templates

Phase 5 — Vessel Architecture (v6.0)

The most architecturally significant evolution. The vessel concept became the foundation of everything.

Multi-Vessel Registry

A single device can hold multiple isolated businesses. Each vessel has its own scoped localStorage — vesselId_cfg, vesselId_inv, vesselId_people, etc. Switching vessels is instantaneous. Data never bleeds between vessels.

Vessel Lineage DNA

Every vessel carries an immutable ancestry chain — who commissioned it, when, from where, and what vessel it descended from. When a steward downloads their data and re-opens it, the lineage is preserved. Across infinite generations of replication, the ancestry chain remains intact.

Bootstrap Migration

Stewards who discover the vessel on the public web can work within the browser, then download their vessel file with all data embedded. The offline sovereign copy opens and restores everything automatically.

Gift-Economy Architecture

Voluntary contributions (TT$25/50/100) are embedded in the vessel. Every contribution is recorded in the vessel's own ledger — the gift economy is itself double-entry.

Platform Awareness

The vessel correctly distinguishes three contexts: file:// (sovereign, offline, unlimited); LAN/private IP (sovereign, network-aware, unlimited); *.github.io (public preview, 3-vessel cap, gift download nudge). The public cap is a natural invitation to download — not a restriction.

The Polish Layer (v6.0 final)

Dirty guard. Every form tracks unsaved changes. Navigating away routes through a confirmation if changes exist. Extended to icon pickers and option grids that bypass standard form field detection.

Payroll accounting correction. The payroll journal entry was posting to wrong accounts (COGS instead of Salaries & Wages). Full account mapping corrected: 6000 Salaries & Wages, 6050 NIS Employer Contribution, 2210/2310/2410/2510 for respective payables. Balance Sheet imbalance resolved.

Tab accessibility. Ten tab bars across the application converted from inline style manipulation to a unified .tab-btn.active class system.

Form validation and messaging. Required field checks, meaningful error messages, and success confirmations across the transaction workflow.

The Depth Pass: v6.1 → v6.3

The v6.0 vessel was architecturally complete. v6.1 through v6.3 represent the precision layer — moving every module from functional to accurate, from capable to instructive, from working to solid.

v6.1 — Caribbean Jurisdiction Depth

Payroll moved from a single-jurisdiction model to six fully accurate Caribbean statutory engines operating simultaneously under one roof. Each jurisdiction was researched and implemented to its current statutory reality:

- **Trinidad & Tobago —** PAYE annualisation, NIS split, Health Surcharge
- **Barbados —** NIS, PAYE, Health Surcharge (HS = 0 per 2024 waiver — present and correct, not absent)
- **Guyana —** NIS, PAYE, scaled rates
- **Jamaica —** NIS, Income Tax, NHT and Education Tax as dedicated deduction lines (not rolled into PAYE)
- **Saint Lucia —** NIC, PAYE
- **Suriname —** SVB, LB, corrected rate structure

When a vessel is commissioned in a given territory, the payroll engine inherits that jurisdiction's statutory rules automatically. Switching jurisdiction via the company profile updates the deduction logic across the HR module.

v6.2 — Vessel-to-Vessel (V2V) Invoice Exchange

The vessel stopped being a closed island and became capable of sovereign commerce without a server. Three exchange channels were implemented under a unified V2V protocol:

- **File channel —** export an encrypted JSON invoice bundle, send via WhatsApp or email, the receiving vessel imports and creates the transaction
- **QR channel —** generate a QR from a live invoice; the receiving vessel scans it from the camera
- **LAN channel —** vessels on the same local network discover each other and exchange invoices directly, peer-to-peer, no internet required

Every imported invoice carries the sender's Vessel Lineage DNA intact. The receiving vessel's vendor record is automatically created or matched, preserving the full ancestry chain across the transaction.

v6.3 — Normalisation and UX Depth Pass

v6.3 is the longest and most granular evolution phase. It addressed six distinct categories simultaneously.

Stock History Foundation

Every stock mutation — initial entry, PO receipt, manual adjustment, invoice dispatch — now appends a dated entry to product.stockHistory. This is the data foundation for the v6.4 EEMD-LASSO-LSTM forecasting layer. Every product carries its complete movement history forward from commissioning.

Import Completeness

importVessel was only restoring ten data keys. Twelve additional modules — employees, payrollRuns, timesheets, punches, deliveries, quotes, credits, campaigns, touchpoints, jobLogs, encounters, recurring — were silently lost on import. A backup exported from a live vessel with a full HR history would return empty on re-import. All twelve now restored.

Mobile — The Depth Pass

v6.3 is the first evolution where mobile is genuinely solid and equal to desktop. Every layout issue was identified and fixed:

- Interface Language, icon picker, currency and tax rate grids: all converted to auto-fill with minmax — wrap naturally on any screen width
- Table overflow: overflow: hidden changed to overflow: visible — this was silently clipping action buttons on all list screens on mobile
- Row action button containers: width: 100% on mobile so buttons never clip regardless of column collapse
- Slide presentation: navigation footer restructured to two rows — dots centred on top, Prev/Next centred below with 110px touch targets
- Global overflow guard: .exp-body * { max-width: 100%; box-sizing: border-box }

Commissioning Wizard — Field Retention and UX

The wizard now retains all entries on back-navigation. Business name is persisted on every keystroke. Returning to any prior step restores all selections and re-enables the Next button correctly. Business structure transfer was also fixed: the wizard stored 'sole' and 'corp' while the profile dropdown expected 'sole_proprietorship' and 'corporation' — the dropdown always showed blank after commissioning. Normalised at source.

Placeholder-as-Teacher

Every blank input across the four main entry screens now carries a contextual placeholder that teaches by showing rather than labelling. Customer name shows real Caribbean business examples. Product cost says "What you pay — used for COGS & margin." Reorder point says "Alert when stock falls to this level." Notes fields added to Customer, Vendor, Invoice, and PO — all fully wired into save, edit, restore, PDF, and preview paths.

Additional v6.3 Fixes

- Advanced Reports: inline toggle with smooth scroll — results open below the card, no more scrolling to the bottom of the page
- Accounting panel render: rAccountingTab() rebuilt to use stable DOM rather than innerHTML surgery that was destroying tab navigation
- Dataset normalisation: test dataset rebuilt from zero with clean accounting state — no pre-baked journal entries causing double-posting
- Scroll-to-top on every screen navigation and wizard step
- Version badge hygiene: stale v5.8 caption in Compound Entry/Exit screen updated to v6.3

By the Numbers

Metric	v5.1	v6.0	v6.3
File size	~180 KB	660 KB	719 KB
Lines of code	~3,200	~11,900	~13,500+
Functional modules	5	16	18+
Accounting	None	Full double-entry	Full double-entry
HR / Payroll	None	Complete (1 jurisdiction)	Complete (6 jurisdictions)
V2V exchange	None	None	3 channels (file, QR, LAN)
Stock history	None	None	Full mutation log
Mobile usability	Basic	Partial	Equal to desktop
Knowledge slides	None	65 slides	65 slides (all current)
Import completeness	N/A	10 data keys	22 data keys
Vessel isolation	None	Full multi-vessel registry	Full multi-vessel registry
Platform awareness	None	file / LAN / public	file / LAN / public
Offline capability	Basic	Full PWA	Full PWA

What Didn't Change

The constraint that shaped everything: one file. No build process. No dependencies served from elsewhere. No server. The file you open is the complete system — not a client connecting to infrastructure.

This constraint is not a limitation. It is the design. It is what makes the vessel portable, sovereign, and eternal. It is what makes it possible to hand someone a file and say: this is yours, forever, no strings.

> *"I am because we are. And we are because I am."*

What Comes Next: v6.4

The v6.3 stock history layer was built specifically as the data foundation for on-device machine learning. Every product now carries 12+ months of dated stock movements — the training set is embedded in the vessel itself.

TF.js EEMD-LASSO-LSTM Forecasting

Target: EEMD (Empirical Mode Decomposition) to isolate trend and seasonal components from stock history, LASSO regression for feature selection, LSTM neural network for sequence prediction — all running inside the vessel via TensorFlow.js. No API call. No cloud. No data leaving the device.

The vessel will learn from its own history and project forward: demand forecasting, cashflow projection, reorder prediction — sovereign, local, continuously improving.

P2P Vessel Mesh

Vessels on the same LAN that can not only exchange invoices (v6.2) but synchronise ledger state, inventory, and HR data peer-to-peer without a server. Conflict resolution architecture already under design.

None of these require changing the core architecture. The vessel is already structured to receive them.

Closing

From v5.1 to v6.3, the vessel grew from a capable tool into something that did not previously exist: a compressed seed carrying a complete business operating system, the knowledge to master it, the mechanism to replicate itself, and the economy to sustain its lineage.

Built in Port of Spain, Trinidad and Tobago. Stewarded for the Caribbean and beyond. MIT licensed — use it, fork it, modify it, distribute it freely.

The vessel is complete. The knowledge is embedded. The sovereignty is yours.

🌱 Sovereignty multiplies. Distribute freely.

Links

Live demo: Business-in-a-Box

Repository: https://github.com/mindful-consumer/bib

Direct download: https://github.com/mindful-consumer/bib/raw/main/index.html

License: MIT